INTEGRAL TRANSFORMS AND REGULARIZED DELTA FUNCTIONS

Force transfer, material measures, and discrete power consistency

Laura A. Miller

Department of Mathematics, University of Arizona

Fluid Dynamics and IBM Theory • Module 1

$$\boldsymbol{F}(\boldsymbol{q},t) \xrightarrow{\text{spread}} \boldsymbol{f}(\boldsymbol{x},t),$$

$$\boldsymbol{u}(\boldsymbol{x},t) \xrightarrow{\text{interpolate}} \boldsymbol{U}(\boldsymbol{q},t).$$

One kernel links two coordinate descriptions.

Claude (Anthropic) was used to assist with formatting, organization, and clarification of the instructor's original notes.

ROADMAP

1. Continuous coupling	2. Material measures	3. Discrete kernels
Reference and spatial coordinates, force spreading, velocity interpolation	The boundary traction term, unified versus partitioned forces, and the Jacobian question	Moment conditions, approximate translation invariance, and the power identity

> The central modeling choice is the measure with respect to which a force density is defined.

TWO COORDINATE DESCRIPTIONS OF ONE SYSTEM

Eulerian description

$\boldsymbol{x} \in \Omega \subset \mathbb{R}^d, \qquad \boldsymbol{u} = \boldsymbol{u}(\boldsymbol{x}, t), \quad p = p(\boldsymbol{x}, t).$

The fluid variables live on a fixed spatial domain.

Lagrangian description

$\boldsymbol{q} \in \mathcal{U} \subset \mathbb{R}^k, \qquad \boldsymbol{x} = \boldsymbol{X}(\boldsymbol{q}, t), \quad k \le d.$

The label $\boldsymbol{q}$ identifies a material point.

> $\mathcal{U}$ is a fixed material-coordinate domain. It may be chosen as a reference configuration, but it need not equal the initial spatial configuration: $\boldsymbol{X}(\boldsymbol{q}, 0) = \boldsymbol{q}$ is not required (for a fiber or surface, $k < d$, it is not even dimensionally possible).

Some texts write s or $\boldsymbol{\xi}$ for the same material coordinate. Module 4 (IBFE) switches to continuum-mechanics notation, $\boldsymbol{\chi}(\boldsymbol{X}, t)$ with $\boldsymbol{X}$ the reference point; the next slide compares the two.

TWO EQUIVALENT LAGRANGIAN NOTATIONS

Classical IB notation (used in this module)

$$\boldsymbol{x} = \boldsymbol{X}(\boldsymbol{q}, t), \qquad \boldsymbol{q} \in \mathcal{U}.$$

The symbol $\boldsymbol{q}$ labels a point on a curve, surface, or material mesh; $\boldsymbol{X}$ is where that point currently *is*.

Continuum-mechanics notation (used from Module 4 on)

$$\boldsymbol{x} = \boldsymbol{\chi}(\boldsymbol{X}, t), \qquad \boldsymbol{X} \in \mathcal{B}_0.$$

Here $\boldsymbol{X}$ is the *reference* point in the reference configuration $\mathcal{B}_0$. This notation makes reference derivatives and stress measures explicit.

These describe the same material-to-spatial map, and the roles of the symbol $\boldsymbol{X}$ are swapped between them. **In this module $\boldsymbol{q}$ is the material label and $\boldsymbol{X}(\boldsymbol{q}, t)$ its current position**; Module 4 (IBFE) switches to $\boldsymbol{\chi}(\boldsymbol{X}, t)$ with $\boldsymbol{X}$ the reference point.

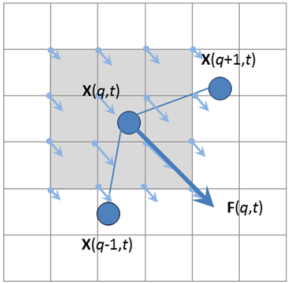
WHAT DOES THE MATERIAL DOMAIN REPRESENT?

- $\mathcal{U}$ is fixed in time and indexes the structural degrees of freedom.
- Integration over $\mathcal{U}$ sums material points, not points of the Eulerian grid.
- Its dimension k may differ from the ambient dimension d: fibers have $k = 1$, surfaces have $k = 2$, and volumetric solids have $k = d$.
- Geometry enters through the current placement $\boldsymbol{X}(\mathcal{U}, t) \subset \Omega$.

$$\boxed{\mathcal{U} \neq \boldsymbol{X}(\mathcal{U}, t) \subset \Omega.}$$

> Calling $\mathcal{U}$ a "reference domain" does not require $\boldsymbol{X}(\boldsymbol{q}, 0) = \boldsymbol{q}$. A convenient material parameterization need not coincide with the initial spatial geometry.

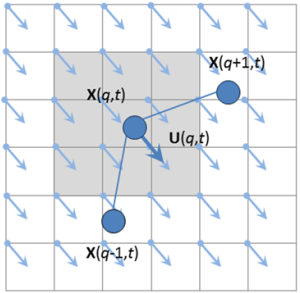
COUPLING GOES IN BOTH DIRECTIONS



Spread force: structure $\longrightarrow$ fluid grid

Figure from deck 1-2.

Schematic: instructor's course notes.



Interpolate velocity: fluid grid $\longrightarrow$ structure

Figure from deck 1-2.

Schematic: instructor's course notes.

The continuum Dirac delta locates the interaction exactly; a discrete kernel shares it among nearby grid points.

THE TWO CONTINUOUS INTERACTION EQUATIONS

Here $\boldsymbol{q}$ is the fixed material label, $\boldsymbol{X}(\boldsymbol{q},t)$ its current position, and $\boldsymbol{F}(\boldsymbol{q},t)$ force per unit material measure $d\boldsymbol{q}$ (Peskin convention, as in decks 1-1, 1-2, 1-4).

Force spreading: Lagrangian → Eulerian

$$\boxed{\boldsymbol{f}(\boldsymbol{x},t)=\int_{\mathcal{U}}\boldsymbol{F}(\boldsymbol{q},t)\,\delta(\boldsymbol{x}-\boldsymbol{X}(\boldsymbol{q},t))\;d\boldsymbol{q}}$$

Velocity interpolation: Eulerian → Lagrangian

$$\boxed{\boldsymbol{U}(\boldsymbol{q},t)=\int_{\Omega}\boldsymbol{u}(\boldsymbol{x},t)\,\delta(\boldsymbol{x}-\boldsymbol{X}(\boldsymbol{q},t))\;d\boldsymbol{x}}$$

For a no-slip material structure, $\partial_t\boldsymbol{X}=\boldsymbol{U}$; with the true Dirac delta this is equivalent to $\partial_t\boldsymbol{X}(\boldsymbol{q},t)=\boldsymbol{u}(\boldsymbol{X}(\boldsymbol{q},t),t)$.

FORCE SPREADING IS A MATERIAL-TO-SPATIAL MAP

Let $\boldsymbol{F}(\boldsymbol{q}, t)\, d\boldsymbol{q}$ be the force carried by a material element. Its Eulerian image is

$$\boxed{\boldsymbol{f}(\boldsymbol{x}, t) = \int_{\mathcal{U}} \boldsymbol{F}(\boldsymbol{q}, t)\delta(\boldsymbol{x} - \boldsymbol{X}(\boldsymbol{q}, t))\; d\boldsymbol{q}.}$$

1. Compute the structural force in material coordinates.
2. Use $\boldsymbol{X}(\boldsymbol{q}, t)$ to locate the material element in physical space.
3. Deposit its force at that location through the Dirac distribution.

> The integral defines a force *measure* on Ω. It remains meaningful when the structure has lower dimension than the fluid domain.

VELOCITY INTERPOLATION CLOSES THE KINEMATIC COUPLING

The reverse transform samples the Eulerian velocity at the current material position:

$$\boldsymbol{U}(\boldsymbol{q},t) = \int_{\Omega} \boldsymbol{u}(\boldsymbol{x},t)\delta(\boldsymbol{x} - \boldsymbol{X}(\boldsymbol{q},t))\; d\boldsymbol{x} = \boldsymbol{u}(\boldsymbol{X}(\boldsymbol{q},t),t)\,.$$

For a no-slip immersed structure,

$$\boxed{\partial_t \boldsymbol{X}(\boldsymbol{q},t) = \boldsymbol{U}(\boldsymbol{q},t).}$$

Dynamic coupling

$\boldsymbol{F} \mapsto \boldsymbol{f}$: the structure loads the fluid.

Kinematic coupling

$\boldsymbol{u} \mapsto \partial_t \boldsymbol{X}$: the fluid moves the structure.

WHAT THE FORCE-SPREADING FORMULA MEANS

The Eulerian force may be a distribution rather than an ordinary function. Its precise meaning is: for every smooth, compactly supported test field $\boldsymbol{\varphi}$,

$$\int_{\Omega} \boldsymbol{f}(\boldsymbol{x},t)\cdot\boldsymbol{\varphi}(\boldsymbol{x})\,d\boldsymbol{x} = \int_{\mathcal{U}} \boldsymbol{F}(\boldsymbol{q},t)\cdot\boldsymbol{\varphi}(\boldsymbol{X}(\boldsymbol{q},t))\;d\boldsymbol{q}.$$

Constant test field

Total Eulerian force equals total Lagrangian force.

Rigid-rotation test field

The first spatial moment gives the corresponding torque identity.

> The delta formula is a distributional push-forward of force measure; it is not a change of integration variables.

TEST FORCE SPREADING ON A SPATIAL REGION: STEPS 1–2

Choose a region $\Gamma \subset \Omega$. Formally, its total Eulerian force is

$$\int_{\Gamma} \boldsymbol{f}(\boldsymbol{x}, t)\, d\boldsymbol{x}.$$

Substitute the spreading relation:

$$\int_{\Gamma} \boldsymbol{f}(\boldsymbol{x}, t)\, d\boldsymbol{x} = \int_{\Gamma} \int_{\mathcal{U}} \boldsymbol{F}(\boldsymbol{q}, t)\, \delta(\boldsymbol{x} - \boldsymbol{X}(\boldsymbol{q}, t))\; d\boldsymbol{q}\, d\boldsymbol{x}.$$

> This notation asks a geometric question: which material points are currently located inside Γ?

TEST FORCE SPREADING ON A SPATIAL REGION: STEPS 3–4

After moving the material integral outside, the inner delta action is

$$\int_{\Gamma} \delta(\boldsymbol{x} - \boldsymbol{X}(\boldsymbol{q}, t))\, d\boldsymbol{x} = \begin{cases} 1, & \boldsymbol{X}(\boldsymbol{q}, t) \in \Gamma, \\ 0, & \boldsymbol{X}(\boldsymbol{q}, t) \notin \Gamma, \end{cases}$$

away from the usual boundary ambiguity.

Thus only material points whose *current* positions lie in Γ contribute.

> The material integration set is the preimage $\boldsymbol{X}_t^{-1}(\Gamma) = \{\boldsymbol{q} \in \mathcal{U} : \boldsymbol{X}(\boldsymbol{q}, t) \in \Gamma\}$.

THE REGION IDENTITY IS THE PHYSICAL PUNCHLINE

Combining the preceding steps gives the formal identity

$$\boxed{\int_{\Gamma} \boldsymbol{f}(\boldsymbol{x}, t)\, d\boldsymbol{x} = \int_{\boldsymbol{X}_t^{-1}(\Gamma)} \boldsymbol{F}(\boldsymbol{q}, t)\, d\boldsymbol{q}.}$$

Interpretation

The total force assigned to a fluid region equals the total force carried by material points occupying that region.

> This statement explains force spreading without imagining a point force at every grid node. The continuum object is a pushed-forward force measure.

CAN WE REALLY SWAP THE INTEGRALS?

True Dirac delta

The delta is a distribution, not an L^1 function. Classical Fubini–Tonelli is therefore not the right justification for the formal double integral.
Define $\boldsymbol{f}$ by its action on smooth test fields and approximate the indicator of Γ.

Regularized kernel δ_h

If δ_h and $\boldsymbol{F}$ are integrable, ordinary Fubini applies. A marker within $O(h)$ of $\partial\Gamma$ then contributes only partially to the region.

> The rigorous continuum starting point is the smooth-test-field identity on the slide “What the force-spreading formula means” (four slides back), not the region identity on the previous slide.

WHERE DID THE JACOBIAN GO?

Reference-measure density

If $\boldsymbol{F}_{\mathrm{ref}}$ is force per $d\boldsymbol{q}$, then

$$\boldsymbol{f}(\boldsymbol{x}) = \int_{\mathcal{U}} \boldsymbol{F}_{\mathrm{ref}}(\boldsymbol{q})\delta(\boldsymbol{x} - \boldsymbol{X}(\boldsymbol{q}))\, d\boldsymbol{q}.$$

No variable substitution is being made.

Current-measure density

For a volumetric map, $d\boldsymbol{x} = J\, d\boldsymbol{q}$, so the same force satisfies

$$\boldsymbol{F}_{\mathrm{ref}} = J\boldsymbol{F}_{\mathrm{cur}}, \qquad J = \det\left(\frac{\partial \boldsymbol{X}}{\partial \boldsymbol{q}}\right).$$

Hence

$$\boldsymbol{f} = \int_{\mathcal{U}} \boldsymbol{F}_{\mathrm{cur}}\delta(\boldsymbol{x} - \boldsymbol{X})J\, d\boldsymbol{q}.$$

No Jacobian is “missing”: it appears only when the chosen force density is expressed per unit current measure.

WHEN DOES A JACOBIAN OR METRIC FACTOR APPEAR?

Suppose the force density is specified per unit *current* geometric measure. Converting it to the fixed material measure introduces the appropriate stretch:

Structure	Measure transformation	Reference/current density relation
Volume, $k = d$	$d\boldsymbol{x} = J\, d\boldsymbol{q}$	$\boldsymbol{F}_{\mathrm{ref}} = J\boldsymbol{F}_{\mathrm{cur}}$
Surface in 3D	$da = J_s\, dA$	$\boldsymbol{F}_{\mathrm{ref}} = J_s\boldsymbol{F}_{\mathrm{cur}}$
Fiber	$ds = \lambda\, dS$	$\boldsymbol{F}_{\mathrm{ref}} = \lambda\boldsymbol{F}_{\mathrm{cur}}$

> The factor belongs to a density conversion. It is not generated merely because $\boldsymbol{X}$ appears inside a delta distribution.

REFERENCE AND CURRENT DENSITIES REPRESENT THE SAME FORCE

For a volumetric placement, the force carried by one material element satisfies

$$\boldsymbol{F}_{\mathrm{ref}}(\boldsymbol{q},t)\,d\boldsymbol{q} = \boldsymbol{F}_{\mathrm{cur}}(\boldsymbol{q},t)\,d\boldsymbol{x}, \qquad d\boldsymbol{x} = J\,d\boldsymbol{q}.$$

Therefore

$$\boxed{\boldsymbol{F}_{\mathrm{ref}} = J\boldsymbol{F}_{\mathrm{cur}}}$$

and the same Eulerian force can be written either way:

$$\boldsymbol{f} = \int_{\mathcal{U}} \boldsymbol{F}_{\mathrm{ref}}\,\delta(\boldsymbol{x}-\boldsymbol{X})\,d\boldsymbol{q} = \int_{\mathcal{U}} \boldsymbol{F}_{\mathrm{cur}}\,\delta(\boldsymbol{x}-\boldsymbol{X})J\,d\boldsymbol{q}.$$

> Always report both the density and its underlying measure; otherwise two equivalent formulas can appear contradictory.

ELASTICITY IS EVALUATED IN MATERIAL COORDINATES

For a volumetric hyperelastic solid, define

$$\boldsymbol{F}_{\mathrm{def}}(\boldsymbol{q},t)=\frac{\partial \boldsymbol{X}}{\partial \boldsymbol{q}}, \qquad J=\det \boldsymbol{F}_{\mathrm{def}}>0.$$

If $W_e(\boldsymbol{F}_{\mathrm{def}})$ is energy per unit reference volume, the first Piola–Kirchhoff stress is

$$\boxed{\boldsymbol{P}_e(\boldsymbol{q},t)=\frac{\partial W_e}{\partial \boldsymbol{F}_{\mathrm{def}}}.}$$

- $\boldsymbol{P}_e$ maps a reference normal to force per unit reference area.
- Its material divergence has units of force per unit reference volume.
- $\boldsymbol{P}_e$ is generally nonsymmetric, even when the Cauchy stress is symmetric.

FROM ELASTIC STRESS TO EULERIAN FORCE

The material stress produces the interior force density

$$\boldsymbol{G} = \mathrm{Div}_{\boldsymbol{q}}\, \boldsymbol{P}_e \qquad \text{in } \mathcal{U},$$

and the solid boundary transmits the reaction $-\boldsymbol{P}_e \boldsymbol{N}$ to the surrounding fluid.

Strong-form elastic force (Griffith–Luo)

$$\boldsymbol{f}_e(\boldsymbol{x}, t) = \int_{\mathcal{U}} \mathrm{Div}_{\boldsymbol{q}}\, \boldsymbol{P}_e(\boldsymbol{q}, t)\, \delta(\boldsymbol{x} - \boldsymbol{X}(\boldsymbol{q}, t))\, d\boldsymbol{q} - \int_{\partial\mathcal{U}} \boldsymbol{P}_e(\boldsymbol{q}, t) \boldsymbol{N}(\boldsymbol{q})\, \delta(\boldsymbol{x} - \boldsymbol{X}(\boldsymbol{q}, t))\, dA(\boldsymbol{q}).$$

The first integral spreads the internal stress divergence; the second spreads the boundary transmission force. The second term may be omitted only when its traction vanishes or when boundary effects are represented another way.

TWO ELASTIC-FORCE CONTRIBUTIONS

Internal force density

$$\boldsymbol{G} = \mathrm{Div}_{\boldsymbol{q}}\, \boldsymbol{P}_e \qquad \text{in } \mathcal{U}.$$

- per unit reference volume
- supported throughout the solid
- regular when $\boldsymbol{P}_e$ is smooth

Transmission force density

$$\boldsymbol{T} = -\boldsymbol{P}_e \boldsymbol{N} \qquad \text{on } \partial\mathcal{U}.$$

- per unit reference area
- supported on the solid boundary
- produces a singular Eulerian force layer

For a lower-dimensional interface, the whole force is singular in the ambient fluid. For a volumetric solid, only the transmitted boundary force is confined to a lower-dimensional set.

UNIFIED WEAK FORCE: ONE MATERIAL DENSITY

Define $\boldsymbol{F}$ through virtual work: for every structural test field $\boldsymbol{V}$,

$$\boxed{\int_{\mathcal{U}} \boldsymbol{F} \cdot \boldsymbol{V}\, d\boldsymbol{q} = -\int_{\mathcal{U}} \boldsymbol{P}_e : \nabla_{\boldsymbol{q}} \boldsymbol{V}\, d\boldsymbol{q}.}$$

The test field need not vanish on $\partial\mathcal{U}$. Integration by parts gives

$$\int_{\mathcal{U}} \boldsymbol{F} \cdot \boldsymbol{V}\, d\boldsymbol{q} = \int_{\mathcal{U}} (\mathrm{Div}\, \boldsymbol{P}_e) \cdot \boldsymbol{V}\, d\boldsymbol{q} - \int_{\partial\mathcal{U}} (\boldsymbol{P}_e \boldsymbol{N}) \cdot \boldsymbol{V}\, dA.$$

> The projected volume field $\boldsymbol{F}$ packages both the interior divergence and the boundary transmission force. No traction-free assumption is being made.

PARTITIONED WEAK FORCE: VOLUME AND BOUNDARY FIELDS

Define the interior density $\boldsymbol{G}$ by

$$\int_{\mathcal{U}} \boldsymbol{G} \cdot \boldsymbol{V}\, d\boldsymbol{q} = -\int_{\mathcal{U}} \boldsymbol{P}_e : \nabla_{\boldsymbol{q}} \boldsymbol{V}\, d\boldsymbol{q} + \int_{\partial\mathcal{U}} (\boldsymbol{P}_e \boldsymbol{N}) \cdot \boldsymbol{V}\, dA,$$

so that $\boldsymbol{G} = \operatorname{Div} \boldsymbol{P}_e$ weakly, and set

$$\boldsymbol{T} = -\boldsymbol{P}_e \boldsymbol{N} \qquad \text{on } \partial\mathcal{U}.$$

Then spread the two contributions separately:

$$\boldsymbol{f} = \int_{\mathcal{U}} \boldsymbol{G}\, \delta(\boldsymbol{x} - \boldsymbol{X})\, d\boldsymbol{q} + \int_{\partial\mathcal{U}} \boldsymbol{T}\, \delta(\boldsymbol{x} - \boldsymbol{X})\, dA.$$

UNIFIED AND PARTITIONED WEAK FORMULATIONS (SIDE-BY-SIDE RECAP)

The previous two slides, placed side by side for comparison; the formulas are the same.

Unified force

Define one density $\boldsymbol{F}$ weakly by

$$\int_{\mathcal{U}} \boldsymbol{F} \cdot \boldsymbol{V}\, d\boldsymbol{q} = -\int_{\mathcal{U}} \boldsymbol{P}_e : \nabla_{\boldsymbol{q}} \boldsymbol{V}\, d\boldsymbol{q},$$

for test fields $\boldsymbol{V}$ that need not vanish on $\partial\mathcal{U}$, and spread

$$\boldsymbol{f} = \int_{\mathcal{U}} \boldsymbol{F}\, \delta(\boldsymbol{x} - \boldsymbol{X})\, d\boldsymbol{q}.$$

Integration by parts shows that $\boldsymbol{F}$ packages both contributions.

Partitioned force

Keep $\boldsymbol{G}$ and $\boldsymbol{T}$ separate:

$$\int_{\mathcal{U}} \boldsymbol{G} \cdot \boldsymbol{V}\, d\boldsymbol{q} = -\int_{\mathcal{U}} \boldsymbol{P}_e : \nabla_{\boldsymbol{q}} \boldsymbol{V}\, d\boldsymbol{q} + \int_{\partial\mathcal{U}} \boldsymbol{P}_e \boldsymbol{N} \cdot \boldsymbol{V}\, dA,$$

$$\boldsymbol{f} = \underbrace{\int_{\mathcal{U}} \boldsymbol{G}\, \delta\, d\boldsymbol{q}}_{\boldsymbol{g}} + \underbrace{\int_{\partial\mathcal{U}} \boldsymbol{T}\, \delta\, dA}_{\boldsymbol{t}}, \qquad \boldsymbol{T} = -\boldsymbol{P}_e \boldsymbol{N}.$$

UNITS CHECK: AN IMMERSED CURVE IN A 2-D FLUID

Let the material label q be arclength along a one-dimensional structure, measured in its reference configuration so that the label is fixed in time, so that $\boldsymbol{X}(q)$ is the current position of that material point, and let the two-dimensional fluid force density be

$$\boldsymbol{f}(\boldsymbol{x}) = \int \boldsymbol{F}(q)\, \delta(\boldsymbol{x} - \boldsymbol{X}(q))\, dq.$$

Then

$$[\boldsymbol{F}] = \frac{\mathrm{N}}{\mathrm{m}}, \qquad [\delta] = \frac{1}{\mathrm{m}^2}, \qquad [dq] = \mathrm{m},$$

and therefore

$$[\boldsymbol{f}] = \frac{\mathrm{N}}{\mathrm{m}} \frac{1}{\mathrm{m}^2} \mathrm{m} = \boxed{\frac{\mathrm{N}}{\mathrm{m}^2}}.$$

> In a 2-D model this is force per unit area. If the 2-D model is read as a slice of an extruded 3-D problem, every force is additionally per unit out-of-plane depth: $[\boldsymbol{F}] = \mathrm{N/m^2}$ and $[\boldsymbol{f}] = \mathrm{N/m^3}$, the familiar 3-D body-force density.

UNITS FOLLOW THE CHOSEN MATERIAL MEASURE

Assume first that the material coordinates have units of length. For a k-dimensional structure in a d-dimensional fluid,

$$[\boldsymbol{F}_{\mathrm{ref}}] = \frac{\mathrm{N}}{\mathrm{m}^k}, \qquad [\delta] = \frac{1}{\mathrm{m}^d}, \qquad [d\boldsymbol{q}] = \mathrm{m}^k,$$

so $[\boldsymbol{f}] = \mathrm{N}/\mathrm{m}^d$.

Structure	k	Reference measure	Force-density units
Fiber	1	length	N/m
Surface	2	area	N/m^2
Volumetric solid	d	d-volume	N/m^d

> If the material labels are dimensionless, $\boldsymbol{F}$ instead has units of force per unit parameter measure; any geometric scale factors are already part of that density or of the quadrature weight.

THE CONTINUUM FORCE CAN BE A SINGULAR LAYER

For a lower-dimensional immersed structure, $\boldsymbol{f}$ is a distribution supported on $\boldsymbol{X}(\mathcal{U}, t)$, not an ordinary pointwise field.

- A curve in 2-D or a surface in 3-D creates a delta-supported force layer.
- Such a layer can produce jumps in pressure or viscous traction across the interface.
- Under classical no-slip coupling, the fluid velocity itself remains continuous across a material interface, although its derivatives may jump.

> “Point force” is misleading here: each material element carries finite force $\boldsymbol{F}\, d\boldsymbol{q}$, and the collection forms a singular measure along the immersed geometry.

FROM A DIRAC DELTA TO A GRID KERNEL

Conventional IB discretizations replace the singular delta distribution by a compactly supported kernel tied to the Eulerian spacing h:

$$\delta_h(\boldsymbol{x}) = \frac{1}{h^d} \prod_{\alpha=1}^{d} \phi\Big(\frac{x_\alpha}{h}\Big) .$$

Why regularize?

- Couple nonmatching grids locally
- Avoid placing a singular distribution on grid nodes
- Obtain sparse interaction operators

What tensor products imply

- Efficient, separable evaluation
- Multidimensional moments inherited from ϕ
- Grid alignment: rotational invariance is only approximate

Sharp-interface and fully variational immersed methods need not use a regularized delta kernel.

DISCRETE SPREADING AND INTERPOLATION

Let $\boldsymbol{x_i}$ be Eulerian grid points, $\boldsymbol{X}_m = \boldsymbol{X}(\boldsymbol{q}_m)$, and w_m Lagrangian quadrature weights.

Matched interaction operators

$$\boldsymbol{f_i} = \sum_m \boldsymbol{F}_m \, \delta_h(\boldsymbol{x_i} - \boldsymbol{X}_m) \, w_m, \qquad \text{(spread)}$$

$$\boldsymbol{U}_m = \sum_{\boldsymbol{i}} \boldsymbol{u_i} \, \delta_h(\boldsymbol{x_i} - \boldsymbol{X}_m) \, h^d, \qquad \text{(interpolate)}$$

> The same kernel is necessary but not, by itself, the whole consistency statement. The Eulerian cell-volume factors and Lagrangian quadrature weights determine the two discrete inner products.

A finite-element IB method may evaluate $\boldsymbol{X}$ and $\boldsymbol{F}$ at element quadrature points rather than only at structural nodes.

WHY DOES THE REGULARIZED KERNEL NEED SPECIAL PROPERTIES?

A generic smooth bump can spread a marker force, but it may not respect the discrete mechanics.

1. Compact support keeps coupling local and sparse.
2. Discrete moments reproduce constants and linear fields.
3. Reflection symmetry avoids a built-in left–right bias.
4. A constant sum of squares reduces sensitivity to subgrid translation.
5. Smoothness controls how interaction values vary as markers cross the grid.

> These are distinct algebraic conditions. Conservation and power consistency must be checked from the paired discrete operators, not inferred from a kernel's appearance.

PROPERTY 1: COMPACT SUPPORT GIVES LOCALITY

For a four-point one-dimensional kernel,

$$\phi(r) = 0 \qquad \text{when } |r| \geq 2.$$

In d dimensions,

$$\delta_h(\boldsymbol{x}) = h^{-d} \prod_{\alpha=1}^{d} \phi(x_\alpha/h),$$

so a Lagrangian sample interacts with only a small tensor-product stencil.

Benefit

Sparse coupling, predictable cost, and communication confined to nearby grid cells.

Tradeoff

Wider support can buy smoother or higher-moment kernels, but increases work and blurs the interface over more cells.

PROPERTY 2: THE ZEROTH MOMENT PRESERVES TOTAL FORCE

Require, for every real offset r,

$$\boxed{\sum_{j\in\mathbb{Z}} \phi(r-j) = 1.}$$

Then spreading a single material force $\boldsymbol{F}_m w_m$ gives

$$\sum_{\boldsymbol{i}} \boldsymbol{f_i} h^d = \boldsymbol{F}_m w_m \prod_{\alpha=1}^{d} \left(\sum_{i_\alpha} \phi(r_\alpha - i_\alpha) \right) = \boldsymbol{F}_m w_m.$$

> The same identity makes interpolation reproduce constant Eulerian velocity fields exactly.

PROPERTY 3: THE FIRST MOMENT CENTERS THE INTERACTION

Require

$$\boxed{\sum_{j\in\mathbb{Z}}(r-j)\phi(r-j)=0.}$$

Equivalently, a marker at $X = rh$ satisfies

$$\sum_{j}(X-jh)\,\delta_h(X-jh)\,h=0.$$

- The discrete force is centered at the marker rather than a nearby grid node.
- Together with the zeroth moment, spreading preserves the corresponding first spatial moment and hence total torque.
- Interpolation reproduces linear fields exactly.

PROPERTY 4: SEPARATE SYMMETRY FROM TRANSLATION INVARIANCE

Even symmetry,

$$\phi(r) = \phi(-r),$$

gives reflection symmetry of the one-dimensional stencil.

Peskin's constant-sum-of-squares condition,

$$\sum_{j \in \mathbb{Z}} \phi(r - j)^2 = C \qquad \text{independent of } r,$$

reduces variation in a marker's self-coupling as it moves through one grid cell.

> Neither condition makes a tensor-product kernel exactly rotationally or translationally invariant. They control specific grid-position effects.

THE FIRST TWO DISCRETE MOMENT CONDITIONS

For every real offset r, impose

$$\sum_{j\in\mathbb{Z}} \phi(r-j) = 1, \qquad \text{zeroth moment,}$$

$$\sum_{j\in\mathbb{Z}} (r-j)\phi(r-j) = 0, \qquad \text{first moment.}$$

Zeroth moment

- interpolates constants exactly
- preserves total force in spreading

First moment

- centers the discrete force
- gives total-torque agreement
- permits exact interpolation of linear fields

Together these identities give $O(h^2)$ interpolation error for sufficiently smooth fields; they do not imply high-order accuracy at a singular fluid–structure interface.

KEEP THE OTHER KERNEL PROPERTIES DISTINCT (RECAP)

Recap of Properties 1 and 4, collected with smoothness so the four non-moment properties can be compared in one place.

Compact support Localizes the *coupling stencil*. The support radius is a kernel choice (for example, two mesh widths for a four-point kernel).

Even symmetry $\phi(r) = \phi(-r)$ gives reflection symmetry. It does not make a tensor-product kernel exactly rotationally invariant.

Sum of squares Peskin's construction can impose

$$\sum_{j \in \mathbb{Z}} \phi(r - j)^2 = C, \qquad C \text{ independent of } r.$$

This reduces variation in self-coupling as a marker moves relative to the grid and is an *approximate translation-invariance* condition.

Smoothness A smoother ϕ produces smoother interpolated quantities, usually at the cost of a wider stencil or additional constraints.

> **Do not confuse the sum-of-squares condition with energy consistency.** Energy consistency comes from discrete adjointness.

WHY THE TWO OPERATORS SHOULD BE ADJOINTS

Define the discrete inner products

$$(\boldsymbol{u}, \boldsymbol{f})_h = \sum_{\boldsymbol{i}} \boldsymbol{u_i} \cdot \boldsymbol{f_i}\, h^d, \qquad (\boldsymbol{U}, \boldsymbol{F})_{\mathcal{U},h} = \sum_m \boldsymbol{U}_m \cdot \boldsymbol{F}_m\, w_m.$$

Substituting the matched spreading formula and exchanging finite sums gives

$$\boxed{(\boldsymbol{u}, \mathcal{S}_h \boldsymbol{F})_h = (\mathcal{J}_h \boldsymbol{u}, \boldsymbol{F})_{\mathcal{U},h}.}$$

> Thus $\mathcal{J}_h = \mathcal{S}_h^*$ with respect to these inner products: the power delivered to the fluid equals the power removed from the structure.

This identity depends on the paired operators, their weights, and evaluation locations. It does *not* require the sum-of-squares condition.

DISCRETE KERNELS USED IN PRACTICE

All of the following start from

$$\delta_h(\boldsymbol{x}) = \frac{1}{h^d} \prod_{\alpha=1}^{d} \phi\Big(\frac{x_\alpha}{h}\Big),$$

but their one-dimensional functions ϕ make different compromises.

- Three-point kernels use a smaller stencil. (The Roma–Peskin–Berger kernel drops Peskin's even–odd condition and is meant for staggered MAC grids.)
- Four-point kernels can combine exact low moments with greater smoothness.
- Simple analytic kernels may satisfy only a subset of Peskin's conditions.
- Wider kernels can target additional moments, smoother derivatives, or improved grid invariance.

> Compare kernels property by property; labels such as "higher order" or "smoother" do not determine performance for every immersed problem.

THREE CLASSICAL ONE-DIMENSIONAL KERNELS

Write M_0 and M_1 for the exact zeroth- and first-moment conditions.

Kernel	Support	M_0	M_1	Even–odd	C in $\sum \phi^2 = C$	Class
Roma–Peskin–Berger 3-point	3	yes	yes	**no**	$C = 1/2$	C^1
Peskin 4-point	4	yes	yes	yes	$C = 3/8$	C^1
Cosine 4-point	4	yes	**no**	yes	$C = 3/8$	C^1

Even–odd: $\sum_{j\,\mathrm{even}} \phi(r-j) = \sum_{j\,\mathrm{odd}} \phi(r-j) = \frac{1}{2}$ (deck 1-2). The three-point kernel drops it and is intended for staggered (MAC) grids (Peskin 2002, Sec. 6; Roma et al. 1999).

For the cosine kernel,

$$\phi_{\cos}(r) = \begin{cases} \dfrac{1}{4}\left(1 + \cos\dfrac{\pi r}{2}\right), & |r| \le 2, \\ 0, & |r| > 2. \end{cases}$$

The cosine kernel does satisfy the constant sum-of-squares identity, but it fails the exact first-moment condition for generic subgrid offsets. These are different tests.

HIGHER-ORDER AND WIDER-SUPPORT KERNELS

Examples include six-point IB kernels (e.g. Bao, Kaye, and Peskin 2016) and B-spline-based families.

Possible benefits

- additional exact moments;
- smoother force and velocity transfer;
- reduced subgrid-position sensitivity;
- improved behavior for selected derivatives.

Possible costs

- a larger interaction stencil;
- more work and parallel communication;
- broader regularization of stress jumps;
- properties that may conflict, such as compactness and smoothness.

> A wider kernel is not automatically more accurate for a solution with a singular interface force.

KERNEL CHOICE IS AN APPLICATION-DEPENDENT TRADEOFF

There is no universally best kernel. A defensible choice reports which properties the application needs.

Numerical questions

- Which moments are reproduced exactly?
- How smooth must interpolated motion and force be?
- Is nonnegativity important?
- What stencil width is affordable?

Resolution questions

- Are marker or quadrature points dense enough relative to h?
- Does the structure remain watertight as it deforms?
- How sensitive are results to subgrid translation and orientation?

> Compare kernels with a problem-relevant convergence and conservation study; a wider or higher-moment kernel is not automatically more accurate near a singular interface.

WHAT HAPPENS UNDER REFINEMENT?

For a normalized family with width proportional to h,

$$\delta_h \longrightarrow \delta \qquad \text{in the sense of distributions as } h \to 0.$$

That statement is about the kernel—not the whole algorithm

Convergence of a complete IB discretization additionally requires:

- consistent Eulerian and Lagrangian spatial discretizations,
- stable time integration and compatible refinement of Δt,
- adequate marker or interaction-quadrature density,
- sufficient solution regularity for the claimed norm and order.

> Regularization spreads an interface force over $O(h)$, so stress jumps are smoothed at finite resolution. Leakage or residual slip need not disappear if the Lagrangian interaction points are under-resolved.

TAKEAWAYS

1. The delta interaction is a push-forward from material measure to spatial measure; it is not a hidden change of variables.
2. A volumetric strong form contains both $\mathrm{Div}\, \boldsymbol{P}_e$ in $\mathcal{U}$ and the transmission traction $-\boldsymbol{P}_e \boldsymbol{N}$ on $\partial\mathcal{U}$.
3. Unified and partitioned weak forms organize those same two physical contributions differently.
4. Zeroth and first moments control total force, torque, and low-order interpolation; constant sum of squares addresses approximate translation invariance.
5. Matched spreading and interpolation should satisfy $\mathcal{J}_h = \mathcal{S}_h^*$, which gives the discrete power identity.

> Always state the measure, kernel properties, and discrete inner products together.

PRIMARY REFERENCES

- C. S. Peskin, “The immersed boundary method,” *Acta Numerica* 11 (2002), 479–517. doi:10.1017/S0962492902000077.
- B. E. Griffith and X. Y. Luo, “Hybrid finite difference/finite element immersed boundary method,” *Int. J. Numer. Meth. Biomed. Eng.* 33 (2017), e2888. doi:10.1002/cnm.2888.
- B. E. Griffith and N. A. Patankar, “Immersed methods for fluid–structure interaction,” *Annu. Rev. Fluid Mech.* 52 (2020), 421–448. doi:10.1146/annurev-fluid-010719-060228.
- A. M. Roma, C. S. Peskin, and M. J. Berger, “An adaptive version of the immersed boundary method,” *J. Comput. Phys.* 153 (1999), 509–534. doi:10.1006/jcph.1999.6293.

Adapted from: spreading/interpolation and kernel postulates after Peskin (2002), Secs. 5–6; the strong and weak elastic-force forms ($\mathrm{Div}\, \boldsymbol{P}_e$ and $-\boldsymbol{P}_e \boldsymbol{N}$) after Griffith and Luo (2017). Additional references at the end of the deck.

APPENDIX: TESTING FORCE ON A SPATIAL REGION

Let $\Gamma \subset \Omega$. A formal indicator-function calculation suggests

$$\int_{\Gamma} \boldsymbol{f}(\boldsymbol{x}, t)\, d\boldsymbol{x} = \int_{\{\boldsymbol{q} \in \mathcal{U}:\, \boldsymbol{X}(\boldsymbol{q}, t) \in \Gamma\}} \boldsymbol{F}(\boldsymbol{q}, t)\, d\boldsymbol{q}.$$

The distributional caveat

The rigorous starting point is the smooth-test-field identity. Approximate the indicator of Γ by smooth functions and pass to the limit. The region identity is unambiguous when the pushed-forward force measure assigns zero mass to $\partial\Gamma$.

> With a regularized δ_h, classical Fubini applies under ordinary integrability assumptions, but a marker within $O(h)$ of $\partial\Gamma$ contributes only partially to the region.

APPENDIX: PESKIN'S FOUR-POINT KERNEL

$$\phi_4(r) = \begin{cases} \frac{1}{8}\left(3 - 2|r| + \sqrt{1 + 4|r| - 4r^2}\right), & 0 \le |r| < 1, \\ \frac{1}{8}\left(5 - 2|r| - \sqrt{-7 + 12|r| - 4r^2}\right), & 1 \le |r| < 2, \\ 0, & |r| \ge 2. \end{cases}$$

- support across four grid points
- zeroth and first moments exact
- $\sum_j \phi_4(r - j)^2 = 3/8$
- nonnegative and C^1

APPENDIX: ROMA–PESKIN–BERGER THREE-POINT KERNEL

$$\phi_3(r) = \begin{cases} \dfrac{1}{3}\left(1 + \sqrt{1 - 3r^2}\right), & |r| < \dfrac{1}{2}, \\ \dfrac{1}{6}\left(5 - 3|r| - \sqrt{-2 + 6|r| - 3r^2}\right), & \dfrac{1}{2} \le |r| < \dfrac{3}{2}, \\ 0, & |r| \ge \dfrac{3}{2}. \end{cases}$$

This C^1, nonnegative kernel satisfies the zeroth and first moment conditions and $\sum_j \phi_3(r-j)^2 = 1/2$, with support across three grid points for generic offsets.

It does *not* satisfy the even–odd condition (at $r = 0$ the even sum is $\phi_3(0) = 2/3$); Peskin (2002, Sec. 6) obtains it by dropping that condition and recommends it only with schemes, such as the staggered MAC scheme, that lack the D^0 null-space problem.

APPENDIX: CHECKING THE COSINE KERNEL

For

$$\phi_{\cos}(r) = \begin{cases} \frac{1}{4}\left(1 + \cos\frac{\pi r}{2}\right), & |r| \le 2, \\ 0, & |r| > 2, \end{cases}$$

direct summation gives, for all real r,

$$\sum_j \phi_{\cos}(r-j) = 1, \qquad \sum_j \phi_{\cos}(r-j)^2 = \frac{3}{8}.$$

However,

$$\sum_j (r-j)\phi_{\cos}(r-j) \neq 0$$

for a generic offset r (computed explicitly on the next slide).

> The example shows why zeroth moment, first moment, and constant sum of squares must be checked separately.

APPENDIX: THE COSINE KERNEL'S FIRST MOMENT, EXPLICITLY

For $0 \le r < 1$ the four contributing offsets are $s = r+1,\ r,\ r-1,\ r-2$. With $\theta = \pi r/2$, the cosines are $-\sin\theta,\ \cos\theta,\ \sin\theta,\ -\cos\theta$, so

$$\sum_s s = 4r - 2, \qquad \sum_s s\cos\frac{\pi s}{2} = 2\cos\theta - 2\sin\theta,$$

and therefore

$$\boxed{M_1(r) = \sum_j (r-j)\phi_{\cos}(r-j) = r - \tfrac{1}{2} + \tfrac{1}{2}\Big(\cos\tfrac{\pi r}{2} - \sin\tfrac{\pi r}{2}\Big).}$$

- M_1 vanishes only at $r = 0,\ \frac{1}{2},\ 1$ and is odd about $r = \frac{1}{2}$.
- Its extremes are $|M_1| \approx 0.021$ (in units of h) at $r \approx 0.21$ and 0.79, where $\sin\theta + \cos\theta = 4/\pi$.
- The same bookkeeping with the squares gives $\frac{1}{16}(4+2) = \frac{3}{8}$, confirming the sum-of-squares value.

QUESTIONS?

Integral transforms and regularized delta functions

REFERENCES (CONTINUED)

- C. S. Peskin, “Numerical analysis of blood flow in the heart,” *J. Comput. Phys.* 25 (1977), 220–252. doi:10.1016/0021-9991(77)90100-0.
- C. S. Peskin (2002), *Acta Numerica* 11, Sec. 6: four-point kernel (eq. 6.27), cosine approximation (eq. 6.28), and the three-point variant; full entry under “Primary references.”
- A. M. Roma, C. S. Peskin, and M. J. Berger (1999): three-point kernel used with the staggered MAC scheme; full entry under “Primary references.”
- F. H. Harlow and J. E. Welch, “Numerical calculation of time-dependent viscous incompressible flow of fluid with free surface,” *Phys. Fluids* 8 (1965), 2182–2189 (MAC grid). doi:10.1063/1.1761178.
- Y. Bao, J. Kaye, and C. S. Peskin, “A Gaussian-like immersed-boundary kernel with three continuous derivatives and improved translational invariance,” *J. Comput. Phys.* 316 (2016), 139–144 (six-point kernel).
- Figures: spreading and interpolation schematics from deck 1-2 (instructor’s course notes).